No.

Date.

Homework #2

习题：1, 3, 5, 7, 8

T1 $P\{X_{n+1}=j \mid X_0=i_0, \cdots, X_{n-1}=i_{n-1}, X_n=i\}$

$$= \begin{cases} \frac{i}{m}\frac{i}{m} & j=i-1 \\ 2\frac{i}{m}\cdot\frac{m-i}{m} & j=i \\ \frac{m-i}{m}\frac{m-i}{m} & j=i+1 \end{cases} = \begin{cases} \frac{i^2}{m^2} & j=i-1 \\ \frac{2i(m-i)}{m^2} & j=i \\ \frac{(m-i)^2}{m^2} & j=i+1 \end{cases}$$

故 $\{X_n\}$ 是时齐马尔可夫链

状态空间 $I=\{0,1,2,\cdots,m\}$

T3 $P\{X_i=1\}=p=1-P\{X_i=0\}$, $0<p<1$

$n\geq 1$, $L_n=\begin{cases} 0 & X_n=0 \\ \max\{1\leq k\leq n : X_nX_{n-1}\cdots X_{n-k+1}=1\} & X_n=1 \end{cases}$

状态空间 $I=\{0,1,2,3,\cdots\}$

$P_{ij}=P\{X_{n+1}=j \mid X_n=i\}$

$$= \begin{cases} p & j=i+1 \\ 1-p & j=0 \end{cases}$$

CS 扫描全能王
3亿人都在用的扫描App

Date.

T5. (1) $P(Y_2=1 \mid Y_0=1, Y_1=6)$

$= P(X_3=1, X_4=1 \mid X_1=1, X_2=1, X_3=6) = 0$

$P(Y_2=1 \mid Y_1=6)$

$= P(X_3=1, X_4=1 \mid \max\{X_2, X_3\}=6) = \frac{1}{66}$

(2) $P(Z_2=12 \mid Z_0=2, Z_1=7)$

$= P(X_3=6, X_4=6 \mid X_1=1, X_2=1, X_3=6) = \frac{1}{6}$

$P(Z_2=12 \mid Z_1=7)$

$= P(X_3=6, X_4=6 \mid X_2+X_3=7) = \frac{1}{36}$

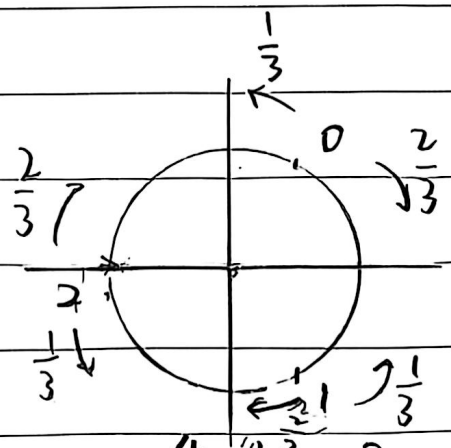
(3) $P(Y_2=1 \mid Y_1=6) \neq P(Y_2=1 \mid Y_0=1, Y_1=6)$

$P(Z_2=12 \mid Z_1=7) \neq P(Z_2=12 \mid Z_0=2, Z_1=7)$

故$\{Y_n\}$, $\{Z_n\}$不具有马尔可夫性

T7 (1)

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{bmatrix} 0 & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & 0 & \frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & 0 \end{bmatrix} \end{array}$$



(2) $P(X_0=0, X_2=0, X_4=1) = P(X_0=0) P_{00}^{(2)} P_{01}^{(2)} = \frac{1}{2} \frac{4}{9} \frac{1}{9} = \frac{2}{81}$

$P(X_2=1) = P(X_0=0) P_{01}^{(2)} + P(X_0=1) P_{11}^{(2)} + P(X_0=2) P_{21}^{(2)}$

$= \frac{1}{2} \times \frac{1}{9} + \frac{1}{4} \times \frac{4}{9} + \frac{1}{4} \times \frac{4}{9} = \frac{5}{18}$

CS 扫描全能王
3亿人都在用的扫描App

$\sum_{n=1}^{\infty} \alpha^n = \frac{1}{1-\alpha}$ $\quad \sum_{n=1}^{\infty} n\alpha^n = \alpha\left[\sum_{n=1}^{\infty}\alpha^n\right]' = \frac{\alpha}{(1-\alpha)^2}$

No.

Date.

T8 (1) $P(X_2=0|X_0=0) = P_{00}^{(2)} = \frac{5}{9}$

$P(X_0=0|X_2=0) = \frac{P(X_0=0, X_1=0)}{P(X_2=0)}$

$= \frac{P(X_0=0)P_{00}^{(2)}}{P(X_0=0)P_{00}^{(2)} + P(X_0=1)P_{10}^{(2)} + P(X_0=2)P_{20}^{(2)}}$

$P^2 = \begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{pmatrix} \frac{5}{9} & \frac{2}{9} & \frac{2}{9} \\ \frac{2}{9} & \frac{7}{9} & 0 \\ \frac{2}{3} & 0 & \frac{1}{3} \end{pmatrix} \end{array}$

$= \frac{\frac{1}{3}\times\frac{5}{9}}{\frac{1}{3}\times\frac{5}{9} + \frac{1}{3}\times\frac{2}{9} + \frac{1}{3}\times\frac{2}{3}} = \frac{5}{13}$

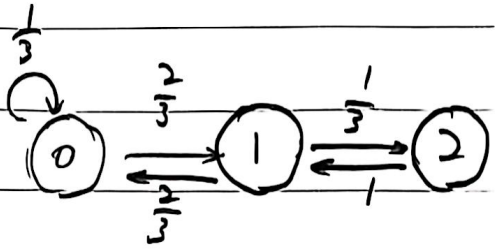
(2) $P(X_1=0) = P(X_0=0)P_{00} + P(X_0=1)P_{10} + P(X_0=2)P_{20} = \frac{1}{3}$

$P(X_1=0, X_3=0, X_4=1, X_6=1) = P(X_1=0)P_{00}^{(2)}P_{01}P_{11}^{(2)} = \frac{70}{729}$

(3) $f_{11}^{(n)} = 0$

$f_{11}^{(2)} = \frac{7}{9}$

$n \geq 3$ 时, $f_{11}^{(n)} = \frac{4}{9}\cdot\left(\frac{1}{3}\right)^{n-2}$



$\mu_1 = \sum_{n=1}^{\infty} n f_{11}^{(n)} = 2\times\frac{7}{9} + \sum_{n=3}^{\infty} n\cdot\frac{4}{9}\cdot\left(\frac{1}{3}\right)^{n-2} =$

$= \frac{14}{9} - \frac{4}{3} - \frac{8}{9} + 4\sum_{n=1}^{\infty} n\left(\frac{1}{3}\right)^n$

$= -\frac{2}{3} + 3 = \frac{7}{3}$

$f_{11} = \frac{7}{9} + \sum_{n=3}^{\infty}\frac{4}{9}\cdot\left(\frac{1}{3}\right)^{n-2} = \frac{7}{9} + \frac{4}{9}\cdot\frac{\frac{1}{3}}{1-\frac{1}{3}} = 1$

CS 扫描全能王
3亿人都在用的扫描App